

TrES-3b

R-2087

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130331 · created 2026-08-10T00:37:10+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456382.9173 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.013
Transit depth (Rp/Rs)^2	3.85 +/- 0.5 [%]
Orbital Inclination (inc)	82.02 +/- 0.38
Airmass coefficient 1 (a1)	1.006 +/- 0.013
Airmass coefficient 2 (a2)	-0.004 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#3 - [514, 90]
Optimal Aperture	3.44
Optimal Annulus	8.21
Transit Duration (day)	0.0602 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.013 vs 0.16309 (deviation 2.53σ)
Duration fit vs published	0.0602 d vs 0.0567 d (deviation 1.01σ)
Mid-time O–C	0.5 min
Depth SNR	15.1
Residual scatter	1.29 %

Light curve

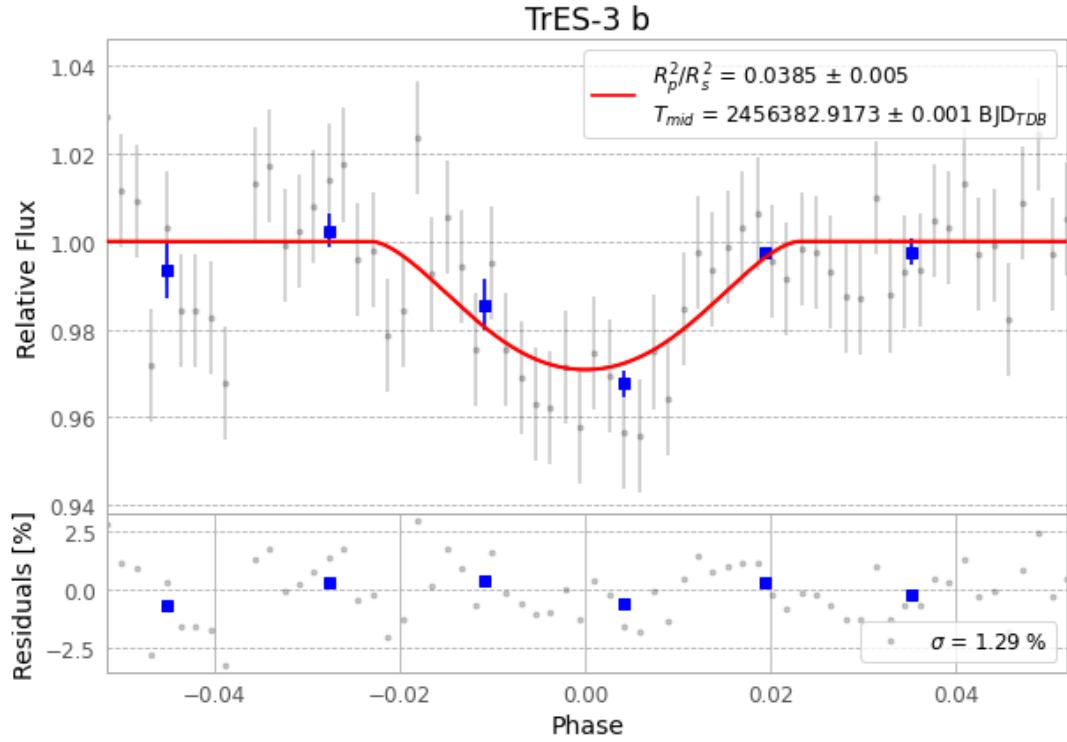


Figure 1 — FinalLightCurve_TrES-3 b_2013-03-31.png (SHA-256 030a30ef3259c6f3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0cfa8d942b551392...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), TRES-3130331082112.FITS → TRES-3130331113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130331.log (500fcc9eaf0a...), FinalLightCurve_TrES-3 b_2013-03-31.png (030a30ef3259...), FinalLightCurve_TrES-3 b_2013-03-31.csv (e21d98490eca...)

Compute resources

Wall time 177.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:37Z.